

Yang Zhou

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EDUCATION

- Carnegie Mellon University** August 2023 - Present
 - Doctorate in Electrical and Computer Engineering; QPA: 3.95/4.00*
 - Advisor:** Professor Beidi Chen
- University of Texas at Austin** August 2019 - May 2023
 - Bachelor of Electrical and Computer Engineering; GPA: 3.99/4.00*
High Honors
 - Graduation Track:** Computer Architecture and Embedded System
 - Courses:** Computer Architecture, ML Hardware Software Co-design, Data Science Lab, Operating Systems, Algorithms, Embedded System Lab

RESEARCH INTERESTS

- Efficient Post-training of Large Language Models, Efficient Inference of Large Language Models, Large Language Model Reasoning, Distributed Training of Large Neural Networks

PUBLICATIONS

- "Sparrow: Sparse Rollout for Stable and Efficient Long-context RL of Large Language Models" **Yang Zhou**, Ranajoy Sadhukhan, Zhaofeng Sun, Zhuoming Chen, Souvik Kundu, Saket Dingliwal, Sai Muralidhar Jayanthi, Aram Galstyan, Haizhong Zheng, Beidi Chen. (2026) preprint
- "Jackpot: Optimal Budgeted Rejection Sampling for Extreme Actor-Policy Mismatch RL" Zhuoming Chen*, Hongyi Liu*, **Yang Zhou***, Haizhong Zheng, Beidi Chen. *Equal Contribution. *ICLR'26*
- "Act Only When It Pays: Efficient Reinforcement Learning for LLM Reasoning via Selective Rollouts" Haizhong Zheng, **Yang Zhou**, Brian R. Bartoldson, Bhavya Kailkhura, Fan Lai, Jiawei Zhao, Beidi Chen. *NeurIPS'25*
- "GSM-Infinite: How Do Your LLMs Behave over Infinitely Increasing Context Length and Reasoning Complexity?" **Yang Zhou***, Hongyi Liu*, Zhuoming Chen, Yuandong Tian, Beidi Chen. *Equal Contribution. Talk link *ICML'25*
- "Sirius: Contextual Sparsity with Correction for Efficient LLM" **Yang Zhou**, Zhuoming Chen, Zhaozhuo Xu, Victoria Lin, Beidi Chen. *NeurIPS'24*
- "LLM Inference Unveiled: Survey and Roofline Model Insights" Zhihang Yuan*, Yuzhang Shang*, **Yang Zhou***, Zhen Dong, Zhe Zhou, Chenhao Xue, Bingzhe Wu, Zhikai Li, Qingyi Gu, Yong Jae Lee, Yan Yan, Beidi Chen, Guangyu Sun, Kurt Keutzer (2024) preprint, cited 228 times (by July 2026)
- "DQRM: Deep Quantized Recommendation Models" **Yang Zhou**, Zhen Dong, Ellick Chan, Dhiraj Kalamkar, Diana Marculescu, Kurt Keutzer (2023) preprint
- "Play It Cool: Dynamic Shifting Prevents Thermal Throttling" **Yang Zhou**, Feng Liang, Ting-wu Chin, Diana Marculescu *DyNN @ ICML'22 (oral)*

INTERNSHIP EXPERIENCE

- Meta Platforms, Inc.** May 2025 - August 2025
 - Research Scientist Intern (on CPT)* Supervisor: Attiano Pontoniere
 - Lead a project studying mid-training effect in downstream reasoning benchmarks. Observe that some models without math reasoning fails to benefit from RL. Also, observe that there exists a sweet spot in mixing thinking trace and math-related web corpus.
 - Explore the topic of synthetically generating a huge sum of puzzle problems and reasoning traces using only a non-reasoning model.

ACADEMIC EXPERIENCE

- Infini-AI-Lab, Carnegie Mellon University** August, 2023 - Present
 - Graduate Student* Supervisor: Professor Beidi Chen
 - LLM efficient post-training with extreme actor-policy mismatch** - Developed Jackpot, a powerful rejection-sampling-based distribution alignment method for establishing extreme actor-policy mismatch, and conducted a comprehensive study enabling block-sparse rollouts with dense training to achieve practical optimal efficiency.
 - LLM reasoning evaluation through synthetic data** - Observe that grade-school-level math problems can be abstracted out into computational graph. Randomly generating computational graph allows for fine-grained control of reasoning complexity and context length. Published in ICML 2025

- **LLM model compression for inference optimization** - Identify the problem of existing Contextual Sparsity techniques in complex reasoning tasks, and solve it by proposing a novel method. Published in NeurIPS 2024
- **Pallas Group, University of California, Berkeley** May 2022 - May 2023
Undergraduate Research Assistant *Supervisor: Professor Kurt Keutzer and Dr. Zhen Dong*
 - Lead a project that optimizes the state-of-the-art recommendation model (DLRM) in both inference and training
 - Compress the size of the state-of-the-art recommendation models by 8X (4-bit quantization) **without performance sacrifice**
 - Combine sparsification and quantization to compress gradient during training to alleviate communication overhead (>99% sparsification, 8-bit quantization)
 - Project code and preprint are open-sourced (Github Repo: Deep Quantized Recommendation Model DQRM)
- **Energy-Aware Computing Group (EnyAC), UT Austin** March 2021 - May 2023
Undergraduate Research Assistant *Supervisor: Professor Diana Marculescu*
 - Lead a project that targets solving Thermal Throttling issues on edge/phone CPUs
 - Observe that edge devices suffer from Thermal Throttling when making continuous ML inference
 - Propose to **dynamically shift** between Dynamic Networks to prevent edge devices from Thermal Throttling
 - Published in 2022 ICML DyNN workshop (oral) see talk here

HONORS & AWARDS

ECE Department Recognition Award for Exemplary Qualifying Exam Performance, Spring 2025	Top 10% Carnegie Mellon University
Carnegie Institute of Technology Dean's Fellowship	Awarded 2023-2024
2022 Distinguished College Scholar	Top 4%, University of Texas at Austin
2021 Distinguished College Scholar	Top 4%, University of Texas at Austin
2023 College Scholar	Top 10%, University of Texas at Austin
Engineering Honor Student	Top 10%, Since October 2020

SKILLS

- **Programming Languages:** Python (Proficient), Java (Familiar), C/C++ (Familiar)
- **Frameworks:** PyTorch (Proficient), Android Studio (Proficient)
- **System Implementation Backend:** CUDA (Familiar), Triton (Familiar)
- **Hardware Languages:** Verilog (Basic), Vitis HLS on C++ (Basic)
- **Languages:** Chinese (Native), English (Proficient)
- **Swimming:** Swam across the Yangzi River in 2016 at the age of 15